

Options — Exam Reference Card

Exam task: Given a graph, you must (1) name the position, (2) explain the payoff logic, (3) give a corporate hedging example.

The 4 Positions at a Glance

Long Call	Buy call	Right to BUY at K	Premium paid	Unlimited	Pay
Short Call	Sell call	Obligation to SELL at K	Unlimited	Premium received	Receive
Long Put	Buy put	Right to SELL at K	Premium paid	K – Premium	Pay
Short Put	Sell put	Obligation to BUY at K	K – Premium	Premium received	Receive

Quick Graph Identification

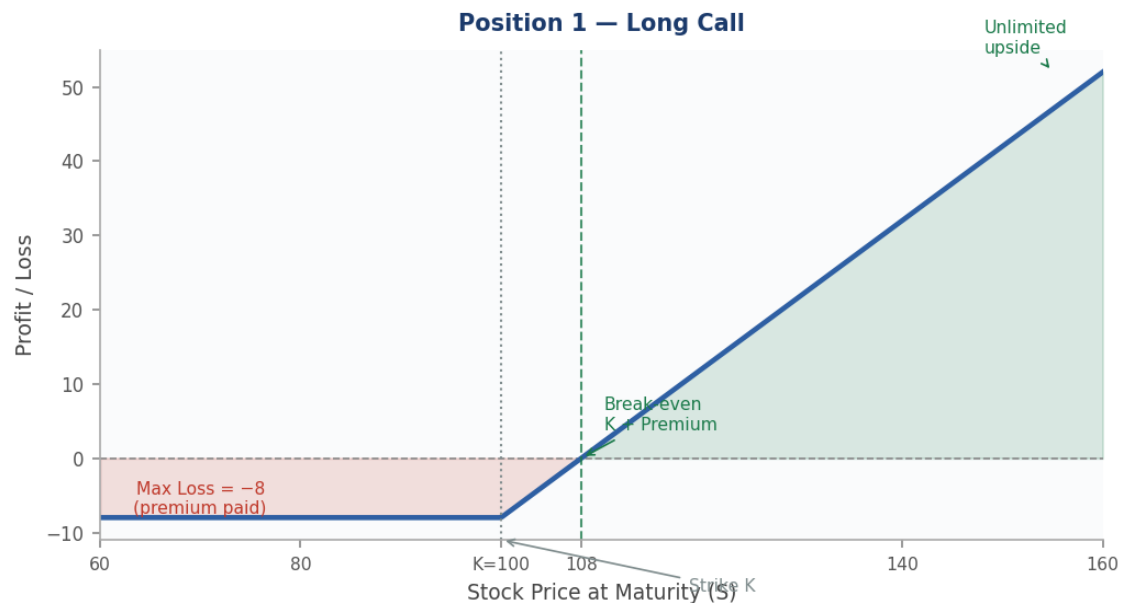
Right side shape	Starting level (y-intercept)	Position
Rising (upward slope)	Below zero (–Premium)	Long Call
Falling (downward slope)	Above zero (+Premium)	Short Call
Flat, left side rises	Below zero (–Premium)	Long Put
Flat, left side falls	Above zero (+Premium)	Short Put

Position 1 — Long Call

What it is: You buy a call option. You pay a premium upfront for the **right to BUY** the asset at strike price K.

Payoff at expiry:

- If $S > K$: **Exercise**. Payoff = $S - K$ | Profit = $(S - K) - \text{Premium}$
- If $S \leq K$: **Do not exercise**. Loss = Premium (maximum possible loss)
- Break-even: $S = K + \text{Premium}$



Corporate use — hedges the risk of buying something that gets more expensive

Importer (FX risk): A Belgian company buying goods priced in USD fears the EUR/USD rate will move against them, making imports costlier. They buy a call on USD — capping their effective exchange rate. If USD stays cheap they let the option expire; if USD spikes the call compensates.

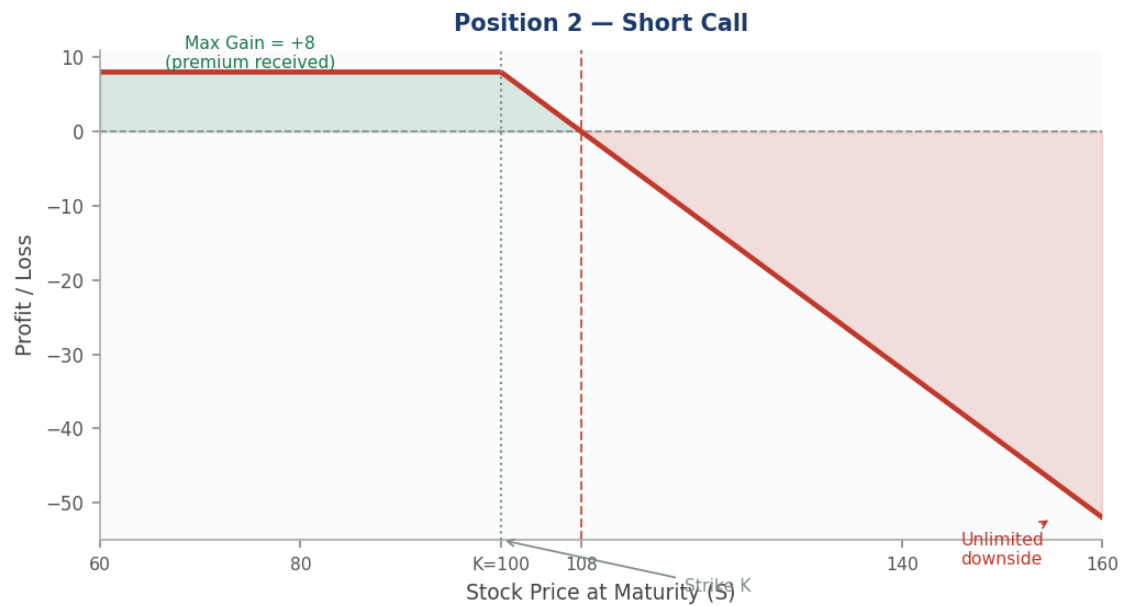
Airline (fuel risk): An airline buying jet fuel buys a call on crude oil — capping fuel costs regardless of how high oil prices go.

Position 2 — Short Call

What it is: You sell a call option. You receive the premium upfront but take on the **obligation to SELL** the asset at K if exercised.

Payoff at expiry:

- If $S \leq K$: **Buyer does not exercise**. Profit = Premium received
- If $S > K$: **Buyer exercises**. You must sell at $K < \text{market price}$. Loss = $(S - K) - \text{Premium}$
- Break-even: $S = K + \text{Premium}$



Corporate use — income generation on assets already owned (covered call)

Covered call: A company holding shares in another firm sells call options on those shares to earn extra income (the premium). They agree to sell at K if the price rises that far. Risk: if the stock surges far above K, they miss the upside.

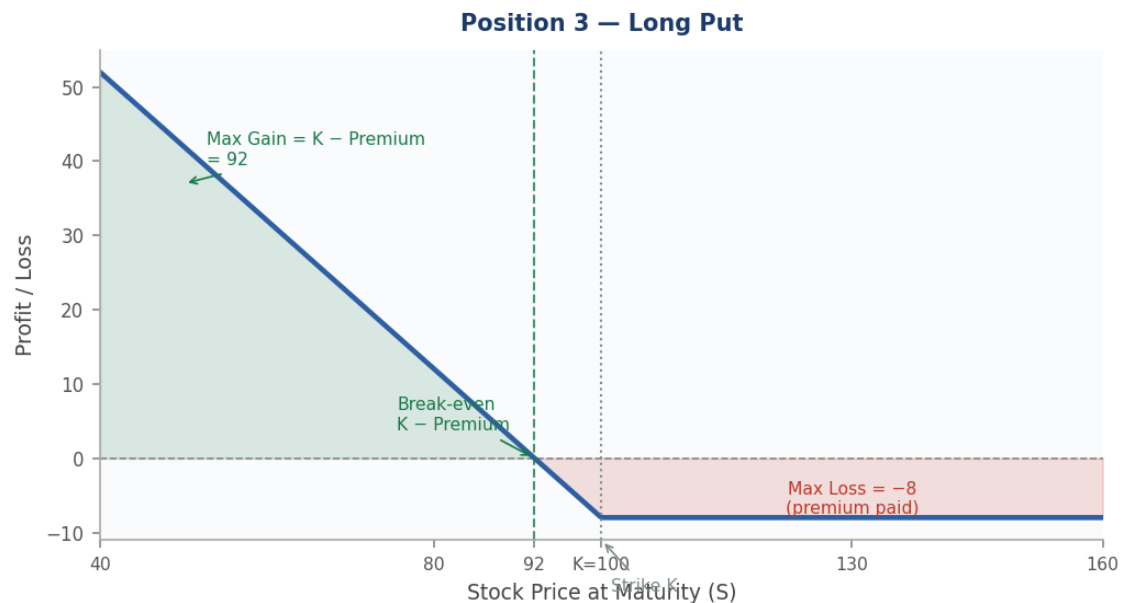
Important — naked short call: Selling a call without owning the underlying is highly speculative (unlimited downside) and is rarely used for corporate hedging.

Position 3 — Long Put

What it is: You buy a put option. You pay a premium upfront for the **right to SELL** the asset at strike price K.

Payoff at expiry:

- If $S < K$: **Exercise**. Payoff = $K - S$ | Profit = $(K - S) - \text{Premium}$
- If $S \geq K$: **Do not exercise**. Loss = Premium (maximum possible loss)
- Break-even: $S = K - \text{Premium}$ | Max Gain = $K - \text{Premium}$ (if $S \rightarrow 0$)



Corporate use — hedges the risk of owning something that loses value (portfolio insurance)

Fund manager: Holds a large equity position and buys put options on it. If the stock crashes, the put pays out, offsetting the loss — classic portfolio insurance.

Belgian farmer: Growing wheat fears prices will fall before harvest. They buy a put on wheat — guaranteeing a minimum selling price (K), while still benefiting if prices rise (they simply do not exercise the put).

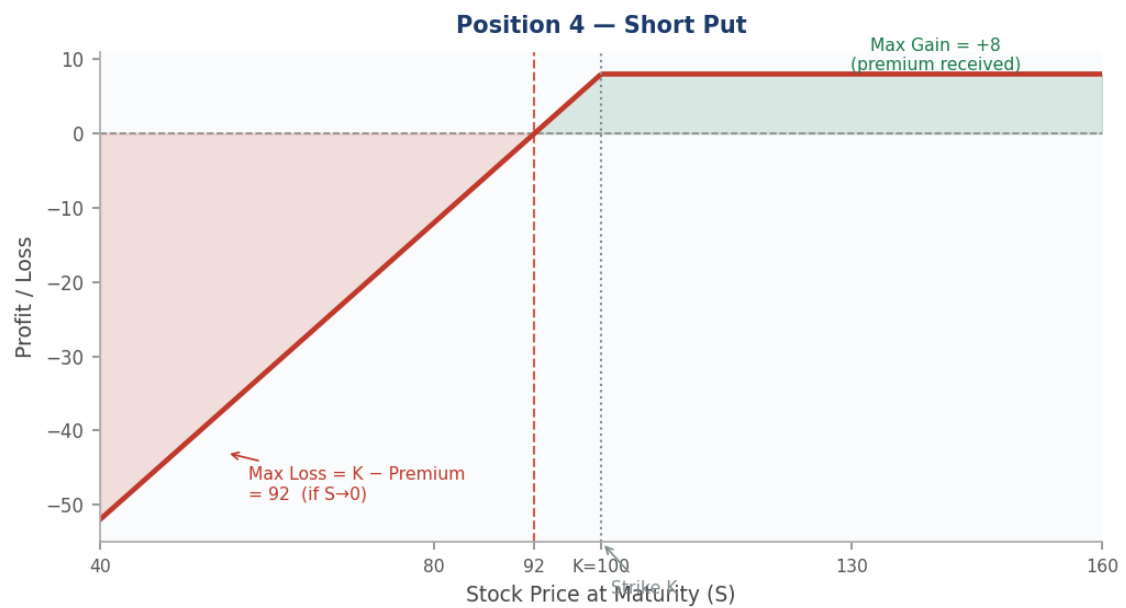
Exporter (FX risk): A company receiving USD in 3 months buys a put on USD — locking in a minimum EUR conversion rate. If USD weakens the put compensates.

Position 4 — Short Put

What it is: You sell a put option. You receive the premium upfront but take on the **obligation to BUY** the asset at K if exercised.

Payoff at expiry:

- If $S \geq K$: **Buyer does not exercise**. Profit = Premium received
- If $S < K$: **Buyer exercises**. You must buy at $K >$ market price. Loss = $(K - S) - \text{Premium}$
- Break-even: $S = K - \text{Premium}$ | Max Loss = $K - \text{Premium}$ (if $S \rightarrow 0$)



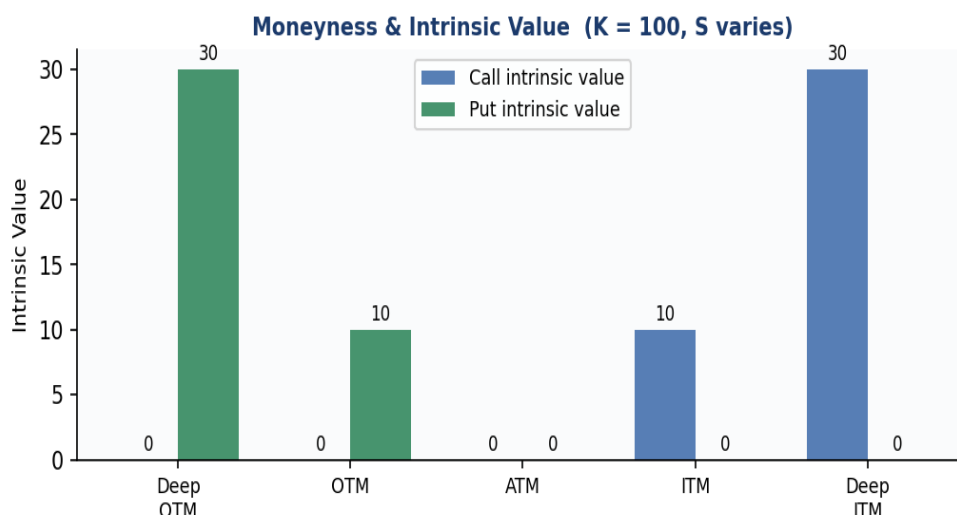
Corporate use — willingness to acquire shares at a lower price

Acquisition strategy: A company wants to buy shares in a target firm but believes the current price is too high. They sell put options at their desired purchase price K . If the stock falls to K , the put is exercised and they acquire the shares at K — exactly the price they wanted. They also pocket the premium as compensation for waiting.

Moneyiness — ITM, ATM, OTM

Moneyiness describes the relationship between the current stock price S and the strike price K . It determines whether immediate exercise would be profitable.

In the Money (ITM)	$S > K$ (exercise is profitable)	$S < K$ (exercise is profitable)	$\max(S-K, 0)$ or $\max(K-S, 0) > 0$
At the Money (ATM)	$S = K$ (indifferent)	$S = K$ (indifferent)	0
Out of the Money (OTM)	$S < K$ (would not exercise)	$S > K$ (would not exercise)	0 (expires worthless)



Option Value = Intrinsic Value + Time Value

- **Intrinsic value** — the immediate exercise value: $\max(S - K, 0)$ for a call; $\max(K - S, 0)$ for a put.
- **Time value** — the extra amount the market pays above intrinsic value, reflecting uncertainty and the possibility that the option moves further in the money before expiry. Time value decays to zero at expiration (theta decay).

Put–Call Parity

Formula: $C - P = S - PV(K) \rightarrow C + PV(K) = P + S$

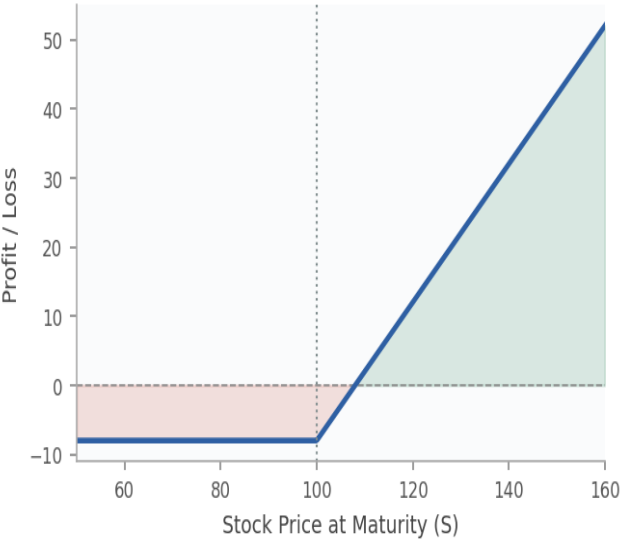
Where: C = call price, P = put price, S = current stock price, $PV(K) = K \cdot e^{-rT}$ (present value of strike).

Put–call parity is a **no-arbitrage relationship** for European options with the same strike K and expiry T on a non-dividend-paying asset. It says two portfolios with identical payoffs must have the same price today:

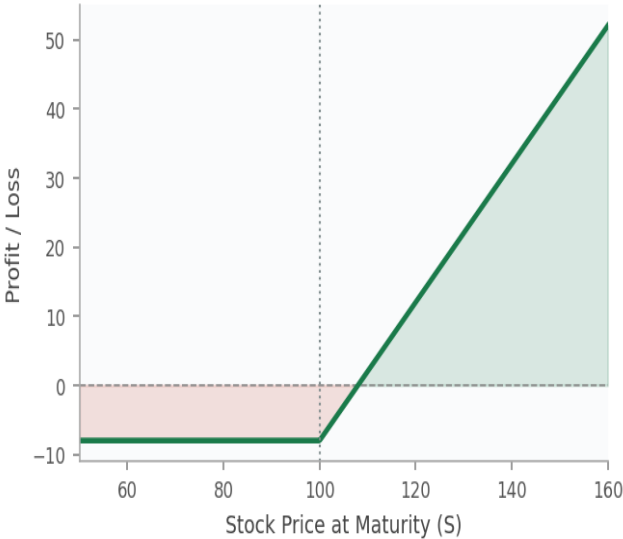
LHS portfolio	RHS portfolio
Long Call + Cash (PV of K) (<i>fiduciary call</i>)	Long Stock + Long Put (<i>protective put</i>)
Payoff at expiry: $S > K$: $(S-K) + K = S$ $S \leq K$: $0 + K = K$	Payoff at expiry: $S > K$: $S + 0 = S$ $S \leq K$: $S + (K-S) = K$

Put-Call Parity: $C - P = S_0 - PV(K)$

LHS: Stock + Long Put



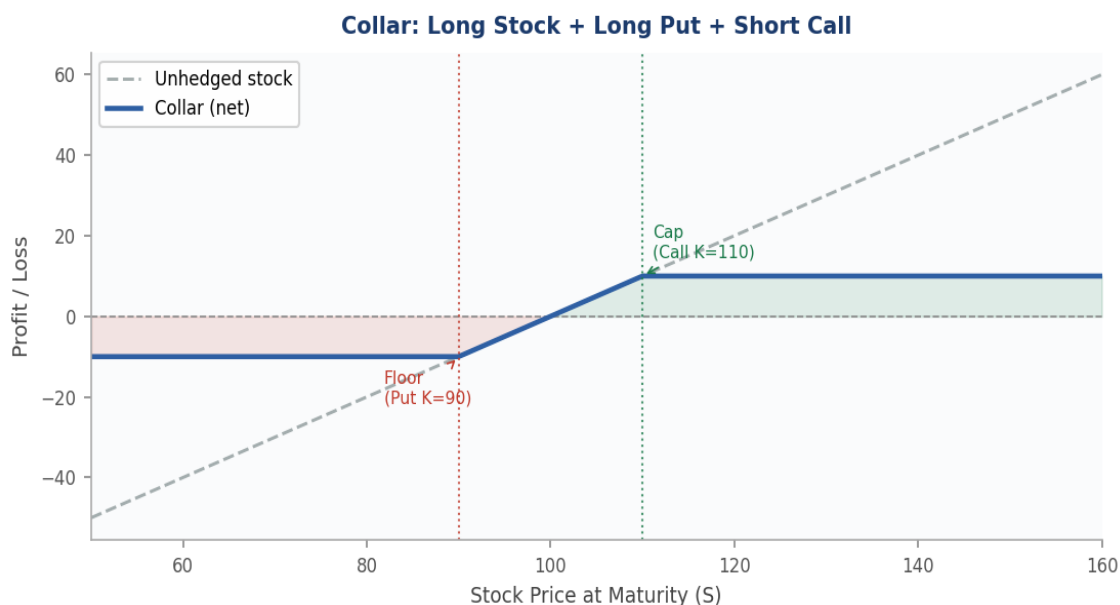
RHS: Long Call + Bond



Collar Strategy (Combined Position)

A **collar** is the most common corporate hedging structure. It combines **three legs**:

1	Already own the underlying asset	Exposed to price movements
2	Buy a put (strike K_{put} , lower)	Sets a floor — limits downside
3	Sell a call (strike K_{call} , higher)	Sets a cap — limits upside; premium offsets cost of put



Zero-cost collar: The premium received from selling the call exactly offsets the premium paid for the put → net cost = 0. The company trades away upside for free downside protection.

Corporate Hedging — One-Line Rules

Price of something you BUY goes UP	Long Call	Caps your maximum purchase cost
Price of something you OWN goes DOWN	Long Put	Floors your minimum selling price
You want income on shares you hold	Short Call	Collect premium; agree to sell at K
You want to buy shares at a lower price	Short Put	Collect premium; agree to buy at K
You own an asset and want to limit both risk and cost	Collar	Buy put (floor) + sell call (cap); can be zero-cost

Key Terminology the Exam Expects

Premium	The price paid (buyer) or received (seller) for the option contract. It is determined at inception and is non-refundable.
Strike price (K)	The pre-agreed price at which the option can be exercised.
In the money (ITM)	Exercising would be immediately profitable: $S > K$ for a call; $S < K$ for a put.
At the money (ATM)	$S \approx K$. Intrinsic value is zero; the option has only time value.
Out of the money (OTM)	Exercising would not be beneficial; the option expires worthless if still OTM at expiry.
Intrinsic value	$\max(S - K, 0)$ for a call; $\max(K - S, 0)$ for a put. The value if exercised right now.
Time value	Option price – intrinsic value. Reflects uncertainty and time remaining. Decays to zero at expiry (theta decay).
Asymmetric payoff	The option holder has a right, not an obligation → bounded maximum loss (the premium), but asymmetric upside/downside.
European option	Can only be exercised at expiry. Used in most textbook formulas (e.g. Black–Scholes).
American option	Can be exercised at any time up to and including expiry. Worth at least as much as a European option.
Put–Call Parity	$C - P = S - PV(K)$. A no-arbitrage relationship between call price, put price, spot price, and present value of the strike (European options only).
Hedging vs. speculation	Corporates use options to REDUCE a pre-existing risk exposure (hedging), not to create new exposure (speculation).
Covered call	Selling a call while already owning the underlying — limits upside but the unlimited loss risk is covered by the shares held.
Protective put	Owning the underlying + buying a put — provides downside insurance while keeping unlimited upside.
Collar	Protective put + covered call on the same underlying — bounds both upside and downside. Can be structured as zero-cost.

Exam Study Tips

Tip 1: Identify graph shape first (flat vs. sloping on each side), then read the sign of the y-intercept. Those two observations uniquely determine all 4 basic positions.

Tip 2: Remember: **Long = pay premium, Short = receive premium**. The starting level of the profit line tells you which side of the trade you are on.

Tip 3: For corporate examples: map the company's UNDERLYING EXPOSURE first, then choose the option that offsets it. An importer has a natural "long USD cost" position → needs a long call on USD to hedge.

Tip 4: Put–call parity can be used to derive any one of C, P, S, PV(K) if the other three are known.

Tip 5: Time value is highest for ATM options and for options with long time to expiry — important for pricing intuition.